Restricted Entropy and Spectrum Properties for the Compressively Sensed Domain in Hyperspectral Imaging

Bernard Lampe, *Student Member, IEEE*, Chein-I Chang, *Life Fellow, IEEE*, Adam Bekit, *Student Member, IEEE,* and Charles Della Porta, *Student Member, IEEE*

***Abstract*— The restricted isometry property (RIP) and restricted conformal property (RCP) are two fundamental properties widely used in compressive sensing (CS) for sampling sparse signals. These two properties are sufficient to preserve the magnitude of a signal vector and the angle between two signal vectors in the compressively sensed band domain (CSBD), respectively. This article derives two new CS properties for hyperspectral signal vectors (HSVs), to be called the restricted entropy property (REP) and restricted spectrum property (RSP), which can be shown to preserve the entropy of an HSV and the spectral similarity between two HSVs in CSBD in correspondence to RIP and RCP, respectively. These two properties enable hyperspectral analysis algorithms to be directly applied to CSBD without loss of data integrity, while the dimensionality of CSBD can be considerably reduced. Most importantly, REP and RSP preserve exploitation algorithm performance without the need for decompression so as to avoid specifying a sparse basis.**

***Index Terms*— Compressive sensing (CS), restricted conformal property (RCP), restricted entropy property (REP), restricted isometry property (RIP), restricted spectrum property (RSP).**

List of Acronyms

CS	Compressive sensing.
HSI	Hyperspectral image.
HSV	hyperspectral signal vector.
RIP	Restricted isometric property.
RCP	Restricted conformal property.
REP	Restricted entropy property.
RSP	Restricted spectrum property.
ODS	Original data space.
CSBD	Compressively sensed band domain.
CSBR	Compressively sensed band rate.
ED	Euclidean distance.
SAM	Spectral angle mapper.
SID	Spectral Information divergence.
SIDAM	Spectral information divergence angle mapper.
HFC	Harsanyi-Farrand-Chang.
VD	Virtual dimensionality.

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Bernard Lampe, Adam Bekit, and Charles Della Porta are with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore, MD 21250 USA.

Chein-I Chang is with the Center of Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore, MD 21250 USA, and also with the Department of Computer Science and Information Management, Providence University, Taichung 02912, Taiwan (e-mail: cchang@umbc.edu).

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I. Introduction

THE RIP of linear algebra has played a key role in developing the field of CS [1]. Concisely, when a matrix satisfies RIP condition for a certain integer sparsity level k, the information of any signal vector with k or fewer nonzero components transformed by the matrix will be preserved allowing for nearly lossless recovery using nonlinear optimization [2]. Using RIP, the theory of CS has shown that using incoherent matrices allows for undersampling sparse signals while preserving the sensed signal energy [3]. As a consequence, nonlinear reconstruction algorithms can be used to recover signals via CS [4]–[6]. Another property derived from RIP, known as RCP, states that the angle between two incoherently sensed, sparse signals is preserved [7], [8]. Generally, any signal vector can be completely specified by the length of the signal which corresponds to the total energy and the angle between the signal vector and the representation basis. A hyperspectral signal is acquired across a range of electromagnetic (EM) wavelengths using hundreds of contiguous spectral bands [9]. RIP and RCP preserve a signal vector length and the angle between two signal vectors in ODS and post-CS acquisition. In other words, all the information about an HSV can be preserved by similar properties in such a way that a compressively sensed signal vector can be reconstructed with high fidelity at very low sample rates by RIP and RCP.

Interestingly, if we interpret the optical spectrum of an HSV in the EM domain as a histogram, it can be normalized to a probability vector [10]. An HSV can be indeed treated as a random variable with the EM wavelength and the normalized spectrum as its sample space and probability distribution, respectively. This normalized spectrum-based probability distribution can be further used to define the entropy of an

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HSV [11]. Using entropy, the discrepancy between probabilistic distributions of two HSVs by SID is developed in [10].

ED, SAM, and SID are ubiquitous measures to quantify the distance and angle between two HSVs [9]. As a result, what the vector length is to a signal vector and the SAM-measured angle is to discrimination between two signal vectors can be directly translated to what entropy is to an HSV and SID-measured spectral similarity is to discrepancy between two HSVs in hyperspectral imaging. With this interpretation, it brings up an interesting conjecture, "Can RIP and RCP be also translated to their counterparts in hyperspectral imaging in CSBD?." This article takes up this issue and derives two new properties, to be called REP which preserves the entropy of an HSV in CSBD after CS and RSP which preserves spectral similarity between two HSVs in CSBD. The novel idea to derive REP is based on the entropy of an HSV that can be used to better describe spectral variability of an HSV than its vector length. It can be shown that such defined entropy for an HSV is preserved in CSBD in a similar manner that the vector length of a signal vector is preserved by RIP. Furthermore, REP can be also used to derive RSP which states that spectral similarity between two HSVs measured by SID can be further preserved in CSBD like the way that the angle between two signal vectors measured by SAM is preserved by RCP after CS. More specifically, REP and RSP are derived as counterparts of RIP and RCP in hyperspectral imaging to preserve entropy of an HSV and spectral similarity between two HSVs in CSBD.

There are several great benefits of using REP and RSP over RIP and RCP. One is to extend deterministic properties (RIP and RCP) to probabilistic properties (REP and RSP) from an information theory point of view. Second, REP and RSP possess better properties to be used for hyperspectral imaging to characterize the spectral variability which is generally caused by many unknown factors such as interfering effects. Third, it has been shown in [10], [12], and [13] that SID is a more effective measure than ED and SAM in hyperspectral data exploitation because SID incorporates both length and direction of a vector into a vector spectrum whose probabilistic profile can faithfully characterize spectral discrimination between two vectors. Fourth and most importantly, the fact that SID-measured probabilistic profiles are preserved in CSBD enables many hyperspectral imaging analysis algorithms to be employed directly in CSBD to produce nearly the same results as they are applied to the original data domain.

II. CS Review

CS is a recent breakthrough in data acquisition and reconstruction of sparse signals, which provides high reconstruction accuracy while data being sampled below the Nyquist rate under the assumption that the signal is sparse in a certain domain and also sensed incoherently [6]. Assume that there are a total number of N HSI pixel vectors acquired by L spectral bands as HSVs, $\{\mathbf{r}_i\}_{i=1}^N$ with $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T \in \Re^L$ being an L-dimensional HSV, where r_{il} is the amount of energy collected in the ith pixel in the lth spectral band. Therefore, we can compress an HSV, $\mathbf{r}_i$, using the following linear representation model:

$$\mathbf{y}_i = \mathbf{\Phi}\mathbf{r}_i + \mathbf{n}_i \tag{1}$$

where $\mathbf{\Phi} \in \Re^{m \times L}$ is the underdetermined, incoherent sensing matrix such that $m \leq L$.

Therefore, according to (1), specifying the sensing matrix $\mathbf{\Phi}$ and sensing dimensionality are required to apply CS to an HSV. In addition, an HSV must also have a sparse representation, denoted by $\mathbf{\Psi}\mathbf{r}_i$ with $\mathbf{\Psi} \in \Re^{L \times L}$ being a sparse basis [14]. However, an optimally sparse basis is hard to find [15]. However, due to universality [16], the sparse basis is only needed for reconstruction which is not necessary for REP [17]. This is because RIP is specified without $\mathbf{\Psi}$ by

$$(1 - \delta_k)||\mathbf{r}||_2^2 \leq ||\mathbf{\Phi}\mathbf{r}||_2^2 \leq (1 + \delta_k)||\mathbf{r}||_2^2 \tag{2}$$

where δ_k is defined as the restricted isometry constant (RIC) and is the smallest possible constant that satisfies (2) for all k-sparse vectors in $\mathbf{r}$. That is, RIP is a sufficient condition for the design of an underdetermined sensing matrix $\mathbf{\Phi}$ [18]. This property ensures that the length of each sampled vector is approximately preserved within the factor $||\mathbf{\Phi}^T\mathbf{\Phi} - \mathbf{I}||_F^2 < \delta_k$ with $||\cdot||_F$ defined as Frobenius norm as $m \to L$.

Another sufficient property of the sensing matrix $\mathbf{\Phi}$ is the RCP which is derived from the RIP as follows [19]:

$$\frac{(1 - \delta_k)}{(1 + \delta_k)} \cos\theta \leq \cos\alpha \leq \frac{(1 + \delta_k)}{(1 - \delta_k)} \cos\theta \tag{3}$$

where $\boldsymbol{\theta}$ is the angle between any two HSVs, $\mathbf{r}_i$ and $\mathbf{r}_j$, and α is the angle between the corresponding pixels in the compressively sensed domain, $\mathbf{y}_i$ and $\mathbf{y}_j$.

Constructing the sensing matrix $\mathbf{\Phi}$ probabilistically is an efficient means to ensure that RIP and RCP conditions are preserved [20]. In addition, using a probability distribution guarantees that the sensing/sampling basis will be incoherent with any structured representation basis. A widely used and well-studied way to generate the entries of $\mathbf{\Phi}$ is to use a Gaussian distribution and normalize the columns explicitly or scale the variance to unit length in expectation [21]. This is a widely used sampling scheme. Under this design, the number of samples, m is specified by

$$m \geq ck \log(L/k) \tag{4}$$

where c is a constant generally considered to be 2 and k is the maximum signal sparsity which can be recovered. This constraint requires estimating the maximum sparsity of the signal, k which remains an open problem [19], [20].

A recent realization has been made on that the HFC method used to estimate VD of HSIs is in fact a sparsity estimation algorithm [22]–[26]. It can be used to choose an optimal sensing/sampling rate for the following experiments which show the results for a range of sensing/sampling rates to illustrate the performance of the HFC method.

The set of HSVs can be sensed using the model described by (1) and each sensed HSV $\{\mathbf{y}_i\}_{i=1}^N$ can be reconstructed individually using nonlinear reconstruction algorithms minimizing the l_1-norm such as orthogonal match pursuit (OMP) [4], basis

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pursuit (BP) [6], and iterative hard thresholding (IHT) [5] to solve the following constrained optimization problem:

$$(\mathbf{\Psi r}_i)^* = \arg\{\min_{\mathbf{\Psi r}_i} ||\mathbf{\Psi r}_i||_1\} \text{ subject to } \mathbf{y}_i = \mathbf{\Phi r}_i. \quad (5)$$

It should be noted that the reconstruction algorithm solving (5) is not required to develop the properties, REP and RSP, as well as hyperspectral data processing algorithms in CSBD. This is because the sparse representation matrix Ψ need not be specified due to universality [1].

An important consequence of the RIP condition in (2) is $||\mathbf{r}_i||_2^2 \approx ||\mathbf{\Phi r}_i||_2^2$ as $m \to L$. A proof of the RIP condition can be found in [1]. However, a statistical treatment for such approximation can be made considerably simpler as follows.

Using (1), the sensing transformation from $\mathbf{r}_i$ to $\mathbf{y}_i$ can be expressed as

$$\mathbf{y}_i = \mathbf{\Phi r}_i = \frac{1}{\sqrt{m}} \begin{bmatrix} \phi_{11} & \cdots & \phi_{1L} \\ \vdots & \ddots & \vdots \\ \phi_{m1} & \cdots & \phi_{mL} \end{bmatrix} \begin{pmatrix} r_{i1} \\ \vdots \\ r_{iL} \end{pmatrix} \quad (6)$$

where $\phi_{ul} \sim N(\mu = 0, \sigma_u^2 = 1)$ for $1 \le u \le m$ is a standard Gaussian distribution and the $1/\sqrt{m}$ term is the column normalization term to define the columns of the sensing matrix as $\mathbf{\Phi} = (\boldsymbol{\phi}_1 \boldsymbol{\phi}_2, \cdots, \boldsymbol{\phi}_L)$, where $\boldsymbol{\phi}_l \in \Re^{m \times 1}$. The expectation of the l_2-norm of the lth column vector $\boldsymbol{\phi}_l$ under the standard Gaussian can be calculated by

$$E[||\boldsymbol{\phi}_l||_2] = \sqrt{||\boldsymbol{\mu}||_2 + \sum_{i=1}^{m} \sigma_i^2} = \sqrt{m}. \quad (7)$$

Therefore, consider the HSV $\mathbf{r}_i$ to be constant and the sensing matrix $\mathbf{\Phi}$ to be random. Then, $\mathbf{y}_i = (y_{i1}, y_{i2}, \ldots, y_{im})^T$ and y_{iu} is distributed Gaussian with $y_{iu} \sim N(\mu = 0, \sigma_u^2 = 1/m \sum_{l=1}^{L} r_{il})$. Knowing the entries of the HSV in the CSBD denoted by $\mathbf{y}_i$ are Gaussian distributed, then the expectation of length of the sensed HSV $||\mathbf{y}_i||_2^2$ can be computed using the same form in (7) as

$$E\left[||\mathbf{y}_i||_2^2\right] = \sqrt{||\boldsymbol{\mu}||_2 + \sum_{i=1}^{m} \left(\frac{1}{m} \sum\nolimits_{l=1}^{L} r_{il}\right)} = \sqrt{\sum_{l=1}^{L} r_{il}^2} = ||\mathbf{r}_i||_2. \quad (8)$$

Therefore, the expected values of the HSV in the ODS and the HSV in the CSBD is $E[||\mathbf{r}_i||_2^2] = E[||\mathbf{y}_i||_2^2] = E[||\mathbf{\Phi r}_i||_2^2]$ and $||\mathbf{r}_i||_2^2 \approx ||\mathbf{y}_i||_2^2 \approx ||\mathbf{\Phi r}_i||_2^2$ as $m \to L$. This result is critical in developing REP and RSP.

Another important result in CS which is useful and will be used in the derivation of REP and RSP is $\mathbf{\Phi}^T \mathbf{\Phi} \to \mathbf{I}$ as $m \to L$, Using (6), we define the columns of the sensing matrix as $\mathbf{\Phi} = (\boldsymbol{\phi}_1 \boldsymbol{\phi}_2, \cdots, \boldsymbol{\phi}_L)$, where $\boldsymbol{\phi}_l \in \Re^{m \times 1}$. Then, the inner product of the sensing matrix results in an $L \times L$ matrix with entries defined by the inner product of the columns $\phi_{uv} = \boldsymbol{\phi}_u^T \boldsymbol{\phi}_v$ with the magnitude of the columns is normalized to one. So, when $u = v$, the inner product becomes 1 and when $u \neq v$ and knowing the vectors are independent identically distributed Gaussian, the inner product is 0, i.e., $\boldsymbol{\phi}_u$ and $\boldsymbol{\phi}_v$ become orthogonal as $m \to L$. Therefore, as dimensionality, m, increases, the diagonal entries approach to 1 and the off-diagonal entries go to zero exponentially.

III. RESTRICTED ENTROPY PROPERTY

The proof showing the preservation of entropy under CS takes a similar form as the proofs for RIP and RCP [5], [16]. For any HSV $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T \in \Re^L$, we follow [9] and [10] and define the spectral information measure (SIM) as the probability measure computed by:

$$p_{il} = \frac{|r_{il}|}{\sum_{l=1}^{L} |r_{il}|}. \quad (9)$$

According to (9), $\mathbf{r}_i$ is considered as a random variable defined on a probability space $X = (\Omega, \Sigma, P)$, where the sample space Ω consisting of $\{r_{il}\}_{l=1}^{L}$ corresponding to all individual wavelengths $\{\lambda_l\}_{l=1}^{L}$ of an HSI, the event space Σ is a power set, and the probability measure P is defined by (9). Using this random HSV, we can simply measure the $p_{il} = p_i(\lambda_l)$ by (9) via SIM. Another way to interpret this probability is to say that p_{il} is the energy from the scene collected in wavelength λ_l.

Furthermore, suppose that $\mathbf{p}_i = (p_{i1}, p_{i2}, \ldots, p_{iL})$ and $\mathbf{q}_i = (q_{i1}, q_{i2}, \ldots, q_{iL})$ are probability vectors calculated by (9) from the ith and jth HSVs. The Kullback–Leibler (KL) divergence between two probability distributions $\mathbf{p}_i$ and $\mathbf{q}_j$ is defined as the difference between the cross entropy

$$D_{\mathrm{KL}}(\mathbf{p}_i||\mathbf{q}_j) = H(\mathbf{p}_i, \mathbf{q}_j) - H(\mathbf{p}_i) \quad (10)$$

where $H(\mathbf{p}_i, \mathbf{q}_j)$ is the joint entropy of $\mathbf{p}_i$ and $\mathbf{q}_j$. Therefore, if we can prove the entropy is preserved under a CS transform $\mathbf{\Phi} \in \Re^{m \times L}$, then the divergence $D_{\mathrm{KL}}(\mathbf{p}_i||\mathbf{q}_j)$ can be also preserved. More specifically, we prove the following theorem

Theorem 1: REP

$$H(\mathbf{p}_i) \to H(\mathbf{\Phi p}_i) \text{ as } m \to L. \quad (11)$$

Proof: (See Appendix A)

Corollary: Let the cross entropy be defined by

$$H(\mathbf{p}_i, \mathbf{q}_j) = -\sum_{l=1}^{L} p_{il} \log q_{jl}. \quad (12)$$

Then

$$H(\mathbf{p}_i, \mathbf{q}_j) \to H(\mathbf{\Phi p}_i, \mathbf{\Phi q}_j) \text{ as } m \to L \quad (13)$$

and

$$D_{\mathrm{KL}}(\mathbf{p}_i||\mathbf{q}_j) + H(\mathbf{p}_i) = D_{\mathrm{KL}}(\mathbf{\Phi p}_i||\mathbf{\Phi q}_j) + H(\mathbf{\Phi p}_i) \text{ as } m \to L. \quad (14)$$

Proof: (See Appendix B)

IV. RESTRICTED SPECTRUM PROPERTY

Using (14), we can further prove the following theorem which validates RSP.

Theorem 2: RSP

$$D_{\mathrm{SID}}(\mathbf{p}_i||\mathbf{q}_j) \to D_{\mathrm{SID}}(\mathbf{\Phi p}_i||\mathbf{\Phi q}_j) \text{ as } m \to L \quad (15)$$

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where SID defined in [9] and [10] is given by

$$D_{\mathrm{SID}}(\mathbf{p}_i, \mathbf{q}_j) = D_{\mathrm{SID}}(\mathbf{q}_j, \mathbf{p}_i) \equiv D(\mathbf{p}_i||\mathbf{q}_j) + D(\mathbf{q}_j||\mathbf{p}_i). \quad (16)$$

Proof: By using $H(\mathbf{p}_i) \rightarrow H(\mathbf{\Phi p}_i)$ in (11) and $H(\mathbf{p}_i, \mathbf{q}_j) \rightarrow H(\mathbf{\Phi p}_i, \mathbf{\Phi q}_j)$ in (13) as $m \rightarrow L$ we can easily prove that as $m \rightarrow L$

$$D_{\mathrm{KL}}(\mathbf{p}_i||\mathbf{q}_j) \rightarrow D_{\mathrm{KL}}(\mathbf{\Phi p}_i||\mathbf{\Phi q}_j) \quad (17)$$

and

$$D_{\mathrm{KL}}(\mathbf{q}_i||\mathbf{p}_j) \rightarrow D_{\mathrm{KL}}(\mathbf{\Phi q}_i||\Phi \mathbf{p}_j). \quad (18)$$

Combining the definition of SID in (16), (17), and (18) yields

$$D_{\mathrm{SID}}(\mathbf{p}_i||\mathbf{q}_j) \approx D_{\mathrm{SID}}(\mathbf{\Phi p}_i||\mathbf{\Phi q}_j). \quad (19)$$

■

Note that the above proof implies that (9) can be further modified in CSBD as

$$p((\mathbf{\Phi r}_i)_l) = p(y_{il}) = \frac{|\langle \mathbf{\Phi e}_l, \mathbf{\Phi r}_i \rangle|}{\sum_{l=1}^{m} |\langle \mathbf{\Phi e}_l, \mathbf{\Phi r}_i \rangle|}. \quad (20)$$

The proof shows that SID is preserved and therefore, SID is also preserved. However, it does not give explicit bounds on the compressed entropy. The assumption made during the proof was the $\cos\theta_l \approx \cos\alpha_l$ for all the angles of $\alpha_1 = \angle(\mathbf{\Phi e}_l, \mathbf{\Phi r}_i)$. This simplification was useful for the proof but did not yield a specific bound. In reality, the angles are bounded as in (3) and each pair of vectors has a slightly different error bound as noted by δ_{kl}. This error bound results from the norm of the difference term in (A3). The sparsity of this vector can vary. We can rewrite the compressed entropy using the error to obtain a lower and upper bound as follows:

$$\frac{\left(\frac{1-\delta_{kl}}{1+\delta_{kl}}\right)|\cos\theta_l|}{\sum_{l=1}^{L}\left(\frac{1-\delta_{kl}}{1+\delta_{kl}}\right)|\cos\theta_l|} \leq \frac{|\cos\alpha_l|}{\sum_{l=1}^{m}|\cos\alpha_l|} \leq \frac{\left(\frac{1+\delta_{kl}}{1-\delta_{kl}}\right)|\cos\theta_l|}{\sum_{l=1}^{L}\left(\frac{1+\delta_{kl}}{1-\delta_{kl}}\right)|\cos\theta_l|}. \quad (21)$$

V. REP and RSP for Hyperspectral Analysis Measures

Hyperspectral measures have found many applications in hyperspectral data analysis [22], [25], [27] such as endmember finding and identification in [28], target detection, linear spectral mixture analysis, and band subset selection [29]. In particular, hyperspectral measures were used to construct a band channel whose capacity can be used to select band subsets [30]. In this section, several commonly used hyperspectral measures, ED, SAM, SID [8]–[10], will be shown to preserve in a CS-compressed domain by RIP, RCP, and REP.

Assume that $\{\mathbf{r}_i\}_{i=1}^{N}$ and $\{\mathbf{y}_i = \mathbf{\Phi r}_i\}_{i=1}^{N}$ are a set of hyperspectral data sample vectors in ODS and their corresponding data sample vectors in CS compressed domain, respectively.

The simplest hyperspectral measure is ED which can be preserved by RIP, i.e.,

$$D_{\mathrm{ED}}(\mathbf{r}_i, \mathbf{r}_j) \approx D_{\mathrm{ED}}(\mathbf{y}_i, \mathbf{y}_j) = ||\mathbf{y}_i - \mathbf{y}_j||_2. \quad (22)$$

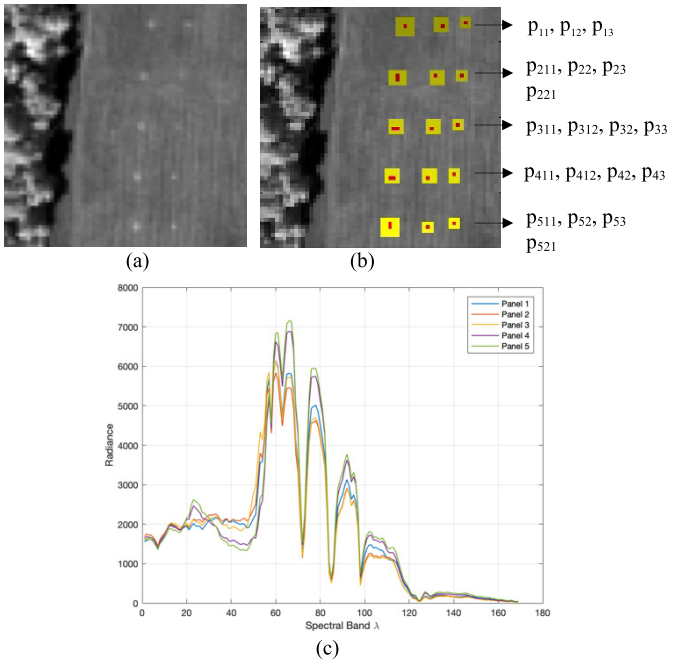


Fig. 1. (a) HYDICE panel scene which contains 15 panels. (b) Ground truth map of spatial locations of the 15 panels. (c) Spectra of five panels obtained by averaging all R panel pixels in each row.

An alternative measure to ED is called SAM which calculates the angle between two data sample vectors that can be preserved by RCP, i.e.,

$$D_{\mathrm{SAM}}(\mathbf{r}_i, \mathbf{r}_j) \approx D_{\mathrm{SAM}}(\mathbf{y}_i, \mathbf{y}_j) = \cos^{-1}\left(\frac{|\langle \mathbf{y}_i, \mathbf{y}_j \rangle|}{||\mathbf{y}_i||_2||\mathbf{y}_j||_2}\right) \quad (23)$$

and the absolute value is taken due to possible negative values caused by two vectors with opposite directions.

Using (16), SID can be preserved by REP as follows:

$$D_{\mathrm{SID}}(\mathbf{r}_j, \mathbf{r}_k) \approx D_{\mathrm{SID}}(\mathbf{y}_i, \mathbf{y}_j) = D(\mathbf{y}_i||\mathbf{y}_j) + D(\mathbf{y}_j||\mathbf{y}_i) \quad (24)$$

where

$$D(\mathbf{y}_i||\mathbf{y}_j) = \sum_{l=1}^{m} p_{il} \log \frac{p_{il}}{p_{jl}} \quad (25)$$

and $p_{il} = |y_{il}| / \sum_{l=1}^{m} |y_{il}|$.

Finally, an interesting measure was developed in [31] which combines SID in (24) and SAM in (23), referred to as SIDAM which can be further preserved by both RCP and REP

$$D_{\mathrm{SIDAM}}(\mathbf{y}_i, \mathbf{y}_j) = D_{\mathrm{SID}}(\mathbf{y}_i, \mathbf{y}_j) \cdot \tan(D_{\mathrm{SAM}}(\mathbf{y}_i, \mathbf{y}_j)). \quad (26)$$

VI. HYDICE Data Experiments

The first image data set is an airborne hyperspectral digital imagery collection experiment (HYDICE) scene shown in Fig. 1(a) which was collected in August 1995 from a flight altitude of 10 000 ft with the ground sampling spectral discrimination measure approximately 1.56 m. It has a size of 64 × 64 pixel vectors shown in Fig. 1(a) along with its ground truth provided in Fig. 1(b), where the center and boundary pixels of objects are highlighted by red and yellow, respectively.

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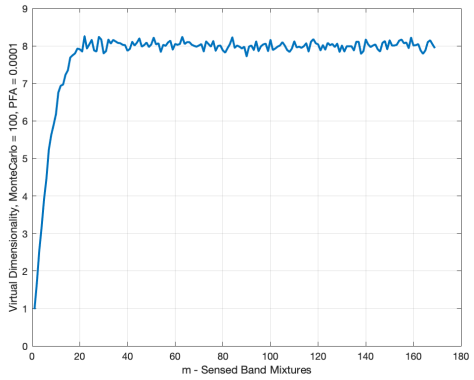


Fig. 2. Sparsity of order, $k \approx 9$ versus number of CSBs, m.

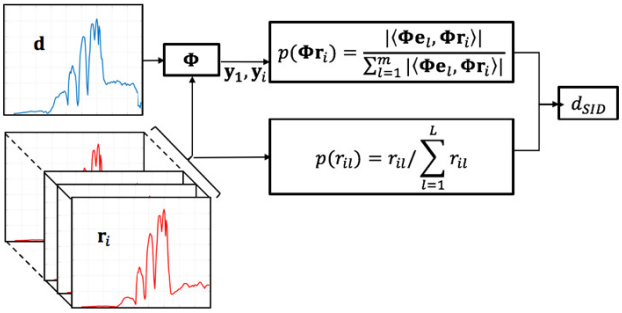


Fig. 3. Experiment 1 setup depicting the comparison of SID between compressively sensed and collected pixels.

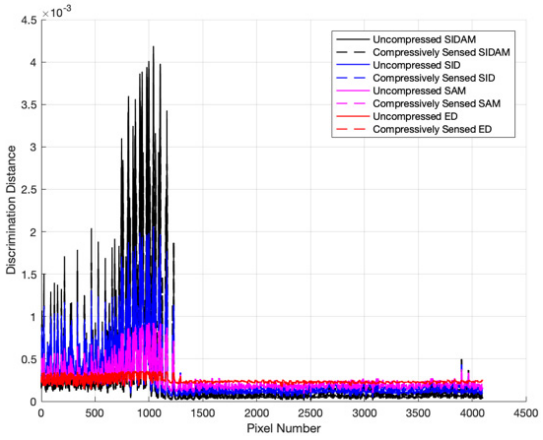


Fig. 4. Discrimination values calculated by $D_j(\mathbf{d}, \mathbf{r}_i)$ and $D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$ for $\mathbf{d} = p_1$ and $\{\mathbf{r}_i\}_{i=1}^{4096}$ and $j =$ ED, SAM, SID, SIDAM.

This scene has been well studied and can be used for experiments, where the center and boundary pixels of objects are highlighted by red and yellow, respectively. Each element in Fig. 1(b) is a square panel and denoted by p_{ij} with rows indexed by $i = 1, 2, \ldots, 5$ and columns indexed by $j = 1, 2, 3$. The 1.56-m-spatial resolution of the image scene suggests that most of the 15 panels are one pixel in size except that the panels in the first column with the second, third, fourth, and fifth rows which are two-pixel panels, denoted by p_{211}, p_{221}, p_{311}, p_{312}, p_{411}, p_{412}, p_{511}, and p_{521}. As a result, there are a total of 19 panel pixels. Fig. 1(b) shows the precise spatial locations of these 19 panel pixels, where red pixels (R pixels) are the panel center pixels and the pixels in yellow (Y pixels) are panel pixels mixed with the background. Fig. 1(c) also plots the signatures of five panels obtained by averaging all R panel pixels in each row. That is, Panel j in Fig. 1(c) is the average value of all R panel pixels in row j. It should be noted that there is an interesting panel pixel p_{212} which is located next to the panel pixel p_{211} horizontally. A detailed description of this particular scene can be found in [8], [22], [25], and [27].

In order to perform practical experiments, we must find a suitable CSBR, m/L. First, we used the HFC method developed in [8] and [21]–[26] to estimate the sparsity of order as VD, $k = 9$ with the false alarm probability P_F set to 10^{-4}. We then increased the number of compressively sensed bands (CSBs), m, up to 169. Fig. 2 shows the convergence of the estimated values of VD at $k \approx 9$ versus m, where $m \approx 20$ seemed to be a reasonably good number of m required for the sparsity of order, $k = 9$.

Experiment 1: In order to validate REP for hyperspectral data analysis, four hyperspectral discrimination measures, D_{ED}, D_{SAM}, D_{SID}, and D_{SIDAM}, were investigated for HYDICE experiments.

Using one of five panel signatures in Fig. 1(c) as a reference signature $\mathbf{d}$, we computed spectral discrimination values between $\mathbf{d}$ and each of all the data sample vectors, $\{\mathbf{r}_i\}_{i=1}^{4096}$, by D_{ED}, D_{SAM}, D_{SID}, and D_{SIDAM} in the ODS and CSBD as shown in Fig. 3, where the spectral discrimination measure is specified by D_{SID}.

Figs. 4–8 plot the spectral discrimination values calculated by D_{ED}, D_{SAM}, D_{SID}, and D_{SIDAM}, between $\mathbf{d} = \mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \mathbf{p}_4, \mathbf{p}_5$, and $\{\mathbf{r}_i\}_{i=1}^{4096}$, respectively, with the number of CSBs, m, was set to 20, where the solid lines and dashed lines are used to specify the results calculated in the original full band domain and CSBD. In addition, Monte Carlo simulations were also used to reduce the variability introduced by the randomly generated sensing matrix $\mathbf{\Phi}$. The experiments were run 100 times and the results were averaged to produce Figs. 4–8, where solid and dashed lines were nearly identical one to another. This indicates that RIP, RCP, and RSP were preserved by ED, SAM, and SID in CSBD, respectively. In addition, both RCP and REP were also preserved by SIDAM in CSBD.

As noted in Figs. 4–8, differences between the dashed and solid lines are not clearly visible. This is because the property of each of 4 measures, ED, SAM, SID, and SIDAM, is preserved in ODS and CSBD resulting from RIP, RCP, REP, and RSP. In other words, the discrimination values of each measure in ODS and CSBD are expected to be very small. As a result, the discrepancy between dashed and solid lines produced by each measure cannot be recognized clearly. It must be this case. Otherwise, if their discrimination values are visibly recognizable, then it implies that the properties of each measure are not preserved. Nevertheless, Figs. 4–8 demonstrate by the plots that each measure did have a different degree in error of property preservation. To further enlarge the differences between solid and dashed lines which are visually undetectable in Figs. 4–8, Figs. 9–13 provide zoom-in

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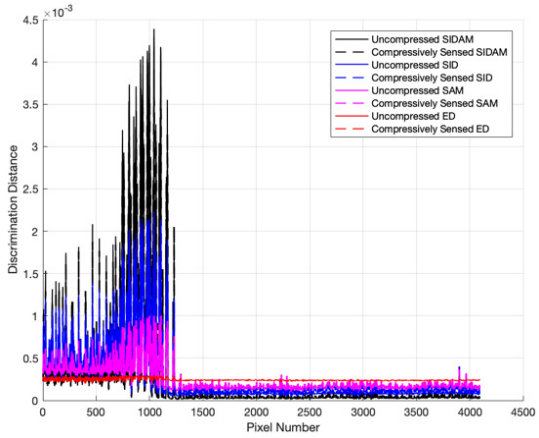


Fig. 5. Discrimination values calculated by $D_j(\mathbf{d}, \mathbf{r}_i)$ and $D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$ for $\mathbf{d} = p_2$ and $\{\mathbf{r}_i\}_{i=1}^{4096}$ and $j =$ ED, SAM, SID, SIDAM.

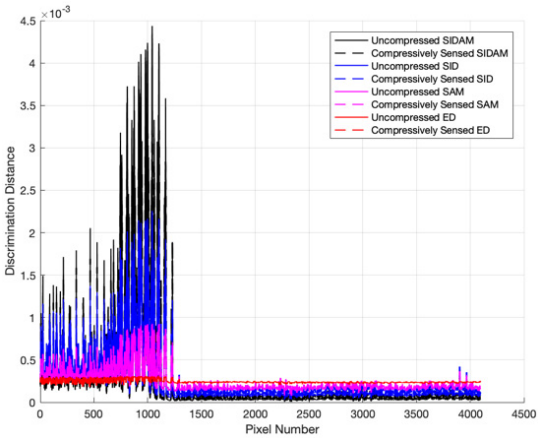


Fig. 6. Discrimination values calculated by $D_j(\mathbf{d}, \mathbf{r}_i)$ and $D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$ for $\mathbf{d} = p_3$ and $\{\mathbf{r}_i\}_{i=1}^{4096}$ and $j =$ ED, SAM, SID, SIDAM.

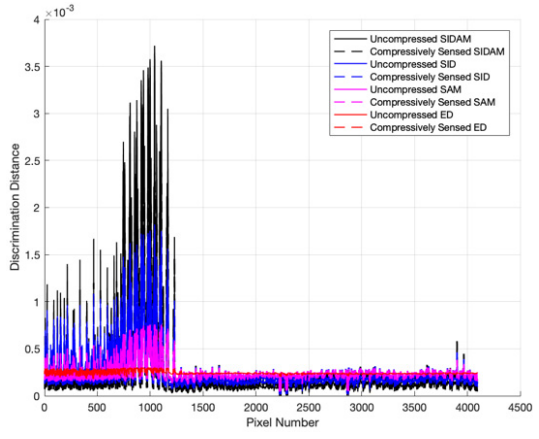


Fig. 7. Discrimination values calculated by $D_j(\mathbf{d}, \mathbf{r}_i)$ and $D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$ for $\mathbf{d} = p_4$ and $\{\mathbf{r}_i\}_{i=1}^{4096}$ and $j =$ ED, SAM, SID, SIDAM.

portions of $D_j(\mathbf{d}, \mathbf{r}_i)$ and $D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$ in Figs. 4–8 of data samples, $\{\mathbf{r}_i\}_{i=2000}^{2050}$ for further inspection.

It is obvious from Figs. 9–13 that both curves of $D_{\mathrm{ED}}(\mathbf{d}, \mathbf{r}_i)$ in the original full band domain and $D_{\mathrm{ED}}(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$ in the

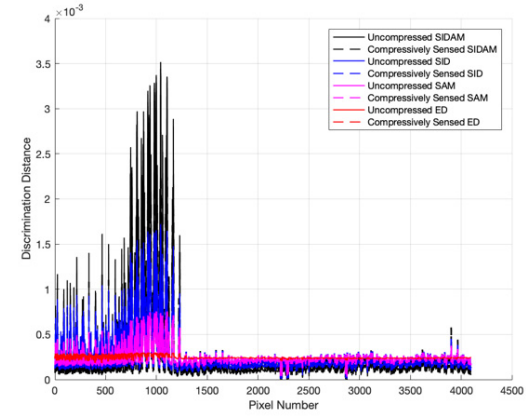


Fig. 8. Discrimination values calculated by $D_j(\mathbf{d}, \mathbf{r}_i)$ and $D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$ for $\mathbf{d} = p_5$ and $\{\mathbf{r}_i\}_{i=1}^{4096}$ and $j =$ ED, SAM, SID, SIDAM.

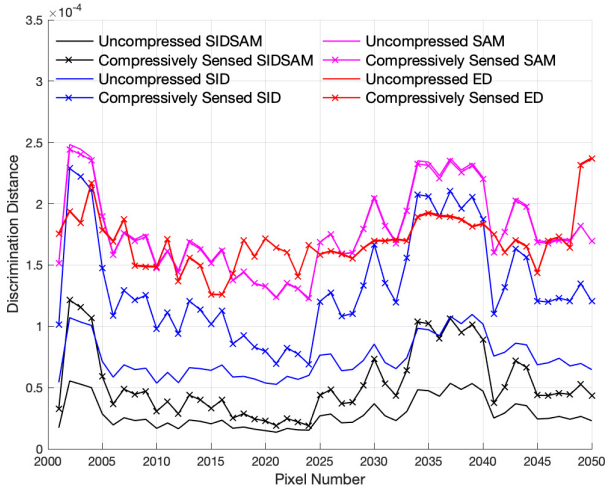


Fig. 9. zoomed-in view portion of Fig. 4 for $\mathbf{d} = p_1$ and $\{\mathbf{r}_i\}_{i=2000}^{2050}$.

CSBD were nearly identical. This indicated that RIP was well preserved by ED with the very tight error bound in (2), $\delta_k \approx 0$, i.e., $1 - \delta_k \approx 1 + \delta_k$. Similarly, it is also very true for RCP which was preserved by SAM with δ_k approximating zero but not as good as the $\delta_k \approx 0$ by ED. However, although the δ_k in the error bound in (2) in Figs. 9–13 produced by SID and SIDAM for REP was also very tight, i.e., δ_k nearly zero, its values were not as good as its counterparts produced by ED and SAM.

To further evaluate the comparisons in Figs. 4–8 quantitatively, we introduce the Pearson correlation coefficient as a criterion to measure the correlation between $\{D_j(\mathbf{d}, \mathbf{r}_i)\}_{i=1}^{N}$ in the original full band domain and $\{D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)\}_{i=1}^{N}$ in the CSBD with $\{\mathbf{r}_i\}_{i=1}^{N}$ being the total number of data sample vectors. Specifically, let $\eta_i^j = D_j(\mathbf{d}, \mathbf{r}_i)$ and $\xi_i^j = D_j(\mathbf{\Phi d}, \mathbf{\Phi r}_i)$. Then, we can calculate $\mu_\eta^j = 1/N \sum_{i=1}^{N} \eta_i^j$ and $\mu_\xi^j = 1/N \sum_{i=1}^{N} \xi_i^j$ which are the means of $\{\eta_i^j\}_{i=1}^{N}$ and $\{\xi_i^j\}_{i=1}^{N}$ along with their corresponding variances, $(\sigma_\eta^j)^2 = 1/N \sum_{i=1}^{N} (\eta_i^j - \mu_\eta^j)^2$. and $(\sigma_\xi^j)^2 = 1/N \sum_{i=1}^{N} (\xi_i^j - \mu_\xi^j)^2$, respectively. Furthermore, let the covariance between $\{\eta_i^j\}_{i=1}^{N}$ and $\{\xi_i^j\}_{i=1}^{N}$ be calculated and denoted by $\mathrm{cov}(\eta^{\mathrm{j}}, \xi^{\mathrm{j}}) = 1/\mathrm{N} \sum_{\mathrm{i}=1}^{\mathrm{N}} (\eta_{\mathrm{i}}^{\mathrm{j}} - \mu_\eta^{\mathrm{j}})(\xi_{\mathrm{i}}^{\mathrm{j}} - \mu_\xi^{\mathrm{j}})$. We can now define the Pearson correlation coefficient between

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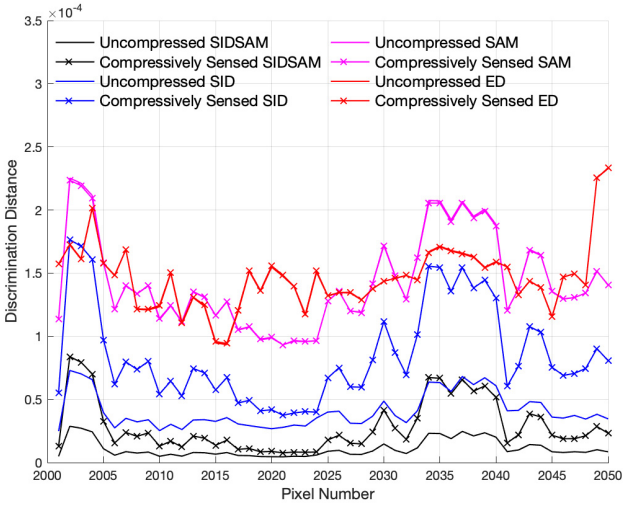


Fig. 10. zoomed-in view portion of Fig. 5 for $\mathbf{d} = p_2$ and $\{\mathbf{r}_i\}_{i=2000}^{2050}$.

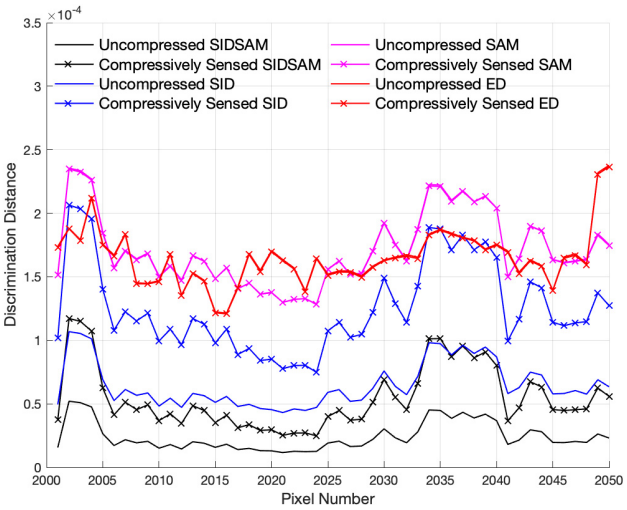


Fig. 11. zoomed-in view portion of Fig. 6 for $\mathbf{d} = p_3$ and $\{\mathbf{r}_i\}_{i=2000}^{2050}$.

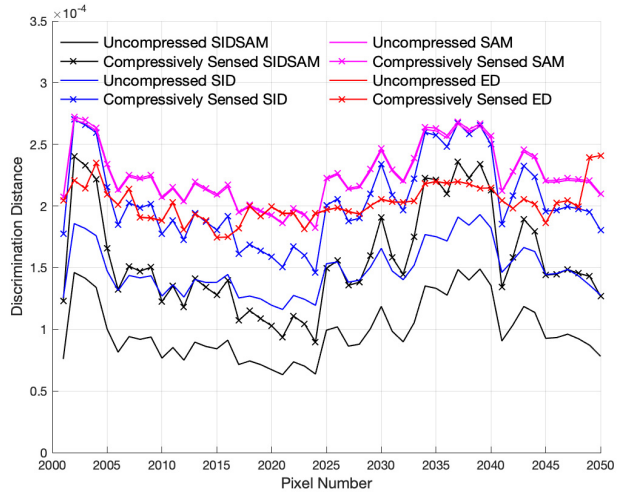


Fig. 12. zoomed-in view portion of Fig. 7 for $\mathbf{d} = p_4$ and $\{\mathbf{r}_i\}_{i=2000}^{2050}$.

$\{\eta_i^j\}_{i=1}^N$ and $\{\xi_i^j\}_{i=1}^N$, denoted by ρ, as

$$\rho = \frac{\mathrm{cov}(\eta, \xi)}{\left(\sigma_\eta^j\right)\left(\sigma_\xi^j\right)} \in [-1, 1] \tag{27}$$

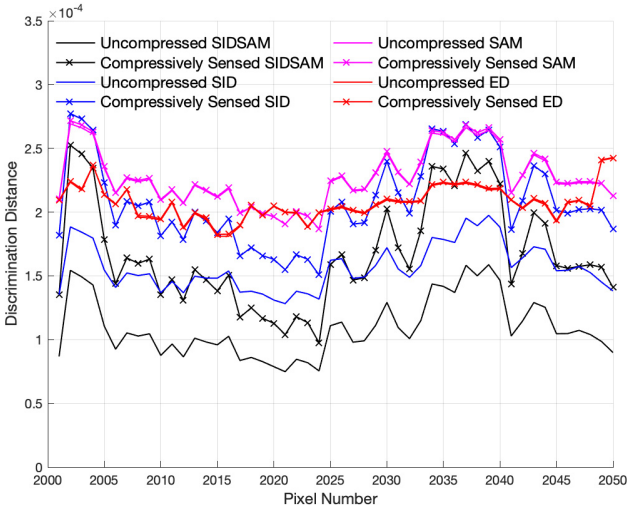


Fig. 13. zoomed-in view portion of Fig. 8 for $\mathbf{d} = p_5$ and $\{\mathbf{r}_i\}_{i=2000}^{2050}$.

TABLE I

ρ IN (27) FOR j = ED, SAM, SID, AND SIDAM, AND $\mathbf{d} = p_1, p_2, p_3, p_4, p_5$, AND $\{\mathbf{r}_i\}_{i=1}^{4096}$

d \ measure	ED	SAM	SID	SIDAM
p_1	1.0000	0.9997	0.9779	0.9887
p_2	0.9998	1.0000	0.9832	0.9908
p_3	1.0000	0.9998	0.9826	0.9906
p_4	0.9999	0.9999	0.9503	0.9841
p_5	1.0000	0.9995	0.9515	0.9828

TABLE II

OVERALL CLASSIFICATION ACCURACY AND THE AVERAGE CLASSIFICATION ACCURACY IN ODS AND CSBD USING D_j = {ED, SAM, SID, SIDAM}

	D_{ED}	D_{SAM}	D_{SID}	D_{SIDAM}
OA - ODS	0.8217	0.9168	0.8482	0.9229
OA - CSBD	0.8081	0.9193	0.8675	0.9186
AA - ODS	0.7708	0.7862	0.7944	0.8393
AA - CSBD	0.7616	0.7904	0.7987	0.8420

which indicates the discrepancy between $\{\eta_i^j\}_{i=1}^N$ and $\{\xi_i^j\}_{i=1}^N$. Table I calculates Pearson correlation coefficient, ρ, in (27) with $N = 4096$ for j = ED, SAM, SID, and SIDAM and $\mathbf{d} = p_1, p_2, p_3, p_4, p_5$, and $\{\mathbf{r}_i\}_{i=1}^{4096}$, where ED was the best followed by SAM, SIDAM, and concluded by SID with the worst. The results in Table I provide very important insights into the error bounds in (2), (3), and (21). Since RIP and RCP are well preserved by (2) and (3), it is expected that their error bounds are very tight and close to zero which indicates that the correlation between data samples in the original full band domain and CSBD is also close to 1.

Also, due to more complicated error bounds specified by (21), SIDAM and SID suffer from larger error bounds. This fact was also reflected in Figs. 4–8 and Table I, where

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TABLE III
INDIVIDUAL CA IN ODS AND CSBD USING $D_j = \{\text{ED, SAM, SID, SIDAM}\}$

	Water	Trees	Asphalt	Bricks	Bitumen	Tiles	Shadows	Meadows	Soil
D_{ED} - ODS	0.9901	0.7794	0.8754	0.7207	0.2792	0.9364	0.7240	0.6359	0.9965
D_{ED} - CSBD	0.9897	0.7843	0.8761	0.7128	0.2672	0.9303	0.7028	0.5959	0.9958
D_{SAM} - ODS	0.9763	0.7917	0.6838	0.8011	0.7278	0.8908	0.8275	0.9541	0.4230
D_{SAM} - CSBD-	0.9795	0.7934	0.7084	0.8037	0.7345	0.8902	0.8280	0.9549	0.4205
D_{SID} - ODS	0.9890	0.8535	0.8961	0.7400	0.3068	0.9363	0.7302	0.7084	0.9892
D_{SID} - CSBD	0.9925	0.7873	0.7314	0.7974	0.5627	0.8795	0.7386	0.7651	0.9340
D_{SIDAM} - ODS	0.9888	0.8351	0.8311	0.7899	0.4304	0.9757	0.7765	0.9367	0.9892
D_{SIDAM} - CSBD	0.9943	0.8003	0.7553	0.8466	0.6183	0.9441	0.7987	0.9003	0.9200

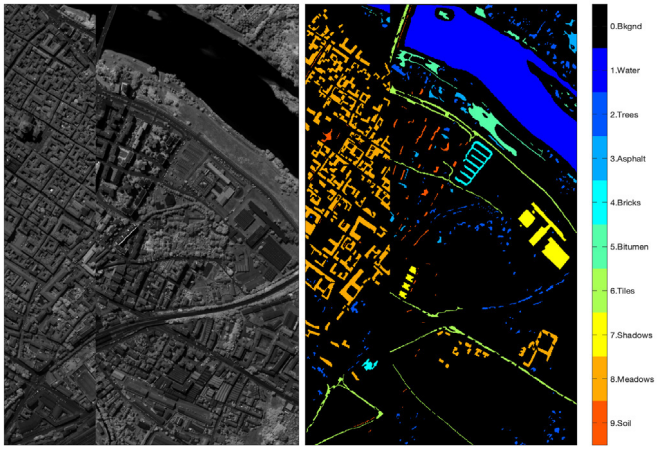


Fig. 14. (a) Rendering of band 80 of the Pavia Center scene. (b) Rendering of the ground truth class labels.

the Pearson correlation coefficients between data samples in the original full band domain and CSBD might not be as close to one as ED and SAM were, their Pearson correlation correlations were still very high close to 0.95. Interestingly, according to Table I, SIDAM resulted in better correlation than SID did because it includes SAM as part of its measure which increased high correlation.

Experiment 2: The second HSI used for the classification experiment is the well-known Pavia Center scene imaged by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor of Pavia University of Northern Italy [31]. The sensor has 115 bands from 430 to 860 nm with spatial resolution of 1.3 and spectral resolution of 4 nm. The image consists of 715 $\times$ 1096 pixel vectors at 102 spectral bands after the removal of the low-SNR water absorption bands. The scene has nine classes and band 80 is rendered in Fig. 14(a) with the ground truth in Fig. 14(b).

The spectral classes consist of water, trees, asphalt, brick, bitumen, tiles, shadows, meadows, and soil. The classifier employed in this experiment utilizes the class means as a descriptor. Each class mean is computed from the image pixels corresponding to the classes in the ground truth image.

Finding the CSBR, m, for this data set is required to perform CS as in (1). Using the HFC method referenced earlier as a sparsity estimation method, the sparsity level is found to be $k = 21$ when the $P_{\text{fa}} = 0.0001$. Using (4), the CSBR is estimated at $m = 67$. Therefore, the classification experiment is run using data in the ODS and CSBD using a CSBR of $m/L = 67/102$.

A nearest centroid classifier is employed based on the class means. Let M be the number of classes in the set $\{C_i\}_{i=1}^{M}$. Each class is a set of HSVs from the HSI such that $\mathbf{r}_i \in C_j$ denotes class membership. The class centroids are computed as the mean of all the pixels in the jth class creating a set of class means $\{\boldsymbol{\mu}_i\}_{i=1}^{M}$. Using this notation, class membership of each HSV in $\{\mathbf{r}_i\}_{i=1}^{N}$ is estimated as

$$\mathbf{r}_i \in C_{i^*} \text{ if } i^* = \arg\{\min_k D_j(\mathbf{r}_i, \mu_k)\} \tag{28}$$

While there are more accurate hyperspectral classifiers [12], [32], this simple classifier is useful to demonstrate the preservation of the HSV discrimination functions $D_j = \{\text{ED, SAM, SID, SIDAM}\}$. The evaluation criteria are the overall classification accuracy, the average accuracy (AA) per class, and the individual class accuracies (CAs) when using data in the ODS and the CSBD. Examining the differences in accuracies in ODS and CSBD demonstrates the preservation properties of RIP, RCP, REP, and RSP in addition to demonstrating utility on an important HSI exploitation algorithm class.

Using the Pavia Center data set, the class overall accuracy (OA) and class AA in the ODS and CSBD is displayed in Table II. This table depicts comparable classification rates using the ODS and data in the CSBD at the CSBR of 67/102.

In addition, the individual CAs are examined to ensure the discrimination metrics are consistent. Table III displays the individual CA for the Pavia Center data set of nine classes in the ODS and CSBD with CSBR of 67/102.

Tables II and III show comparable OA, AA, and CA using data in the ODS and CSBD. While the results may not be identical, they indeed support the derivations in Sections III and IV. In addition, this result also shows the utility of using RIP, RCP, and RSP for hyperspectral discrimination measures defined in Section V.

VII. CONCLUSION

In this article, we coined and proved two new properties of CS for hyperspectral imaging, REP and RSP, both of which have not been explored in the past. It shows that the entropy of an HSV can be preserved in the CSBD. With the help of REP, it further shows that SID between two HSVs can be also preserved in the CSBD. As a result of preserving SID in the

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CSBD, RSP is also validated. Since RSP preserves not only ED between two data samples but also their angle, RSP can be considered as a generalization of RIP and RCP. In other words, what ED is to RIP and what SAM is to RCP are also applied to what entropy is to REP and what SID is to RSP, respectively. Finally, experiments conducted based on real HSIs show that all the four properties, RIP, RCP, REP, and RSP, are preserved in the CSBD.

Appendix A
Proof of Theorem 1

The first step of the proof is to express the entropy in terms of the angles between the basis vectors and the pixel vector $\mathbf{r}_i$ which is denoted by $\theta_l = \angle(\mathbf{e}_l, \mathbf{r}_i)$, where the $\mathbf{e}_l = (0, \ldots, 0, \underbrace{1}_{l}, 0, \ldots, 0)^T \in \Re^L$ is the basis vector with the lth position equal to 1 and all other positions equal to 0

$$\begin{aligned} H(\mathbf{p}_i) &= -\sum_{l=1}^{L} p_{il} \log p_{il} = -\sum_{l=1}^{L} \frac{|r_{il}|}{\bar{r}_i} \log \frac{|r_{il}|}{\bar{r}_i} \\ &= -\sum_{l=1}^{L} \left(|\langle \mathbf{e}_l, \mathbf{r}_i \rangle| / \sum_{l=1}^{L} |\langle \mathbf{e}_l, \mathbf{r}_i \rangle| \right) \\ &\quad \times \log \left(|\langle \mathbf{e}_l, \mathbf{r}_i \rangle| / \sum_{l=1}^{L} |\langle \mathbf{e}_l, \mathbf{r}_i \rangle| \right) \\ &= -\sum_{l=1}^{L} \left(|\mathbf{e}_l^T \mathbf{r}_i| / \sum_{l=1}^{L} |\mathbf{e}_l^T \mathbf{r}_i| \right) \log \left(|\mathbf{e}_l^T \mathbf{r}_i| / \sum_{l=1}^{L} |\mathbf{e}_l^T \mathbf{r}_i| \right) \end{aligned} \tag{A1}$$

where $\bar{r}_i = \sum_{l=1}^{L} |r_{il}|$. Furthermore, (A1) can be reexpressed as

$$\begin{aligned} H(\mathbf{p}_i) &= -\sum_{l=1}^{L} \left(||\mathbf{e}_l|| ||\mathbf{r}_i|| \cos\theta_l | / \sum_{l=1}^{L} ||\mathbf{e}_l|| ||\mathbf{r}_i|| \cos\theta_l | \right) \\ &\quad \times \log \left(||\mathbf{e}_l|| ||\mathbf{r}_i|| \cos\theta_l | / \sum_{l=1}^{L} |\cos\theta_l| \right) \\ &= -\sum_{l=1}^{L} \left(|\cos\theta_l| / \sum_{l=1}^{L} |\cos\theta_l| \right) \log \left(|\cos\theta_l| / \sum_{l=1}^{L} |\cos\theta_l| \right) \end{aligned} \tag{A2}$$

where the norm $||\cdot||$ is the l_2-norm, $||\cdot||_2$. It should be noted that the magnitudes of $||\mathbf{r}_i||$ and $||\mathbf{e}_l||$ are constant. So, they are cancelled in (A1).

Now we can express $\cos\theta_l$ as

$$\cos\theta_l = \frac{||\mathbf{e}_l||_2^2 - ||\mathbf{r}_i||_2^2 - ||\mathbf{e}_l - \mathbf{r}_i||_2^2}{2||\mathbf{e}_l||_2^2 ||\mathbf{r}_i||_2^2}. \tag{A3}$$

By taking advantage of the RIP condition in (2), then $||\mathbf{r}_i||_2^2 \approx ||\mathbf{\Phi}\mathbf{r}_i||_2^2$ and we can replace the magnitudes in the law of cosines with the transformed magnitudes

$$\cos\theta_l \approx \cos\alpha_l = \frac{||\mathbf{\Phi}\mathbf{e}_l||_2^2 - ||\mathbf{\Phi}\mathbf{r}_i||_2^2 - ||\mathbf{\Phi}\mathbf{e}_l - \mathbf{\Phi}\mathbf{r}_i||_2^2}{2||\mathbf{\Phi}\mathbf{e}_l||_2^2 ||\mathbf{\Phi}\mathbf{r}_i||_2^2}. \tag{A4}$$

Now we redefine the entropy of $\mathbf{\Phi}\mathbf{r}_i$, $H(\mathbf{\Phi}\mathbf{r}_i)$, in the compressed domain via Φ with $\alpha_l = \angle(\Phi\mathbf{e}_l, \Phi\mathbf{r}_i)$. In addition, this redefinition allows the angle to be preserved by the RCP in (3)

$$\begin{aligned} &H(\mathbf{\Phi}\mathbf{r}_i) \\ &= -\sum_{l=1}^{m} \left(||\Phi\mathbf{e}_l|| ||\Phi\mathbf{r}_i|| \cos\alpha_l | / \sum_{l=1}^{m} ||\mathbf{\Phi}\mathbf{e}_l|| ||\mathbf{\Phi}\mathbf{r}_i|| \cos\alpha_l | \right) \\ &\quad \times \log \left(||\mathbf{\Phi}\mathbf{e}_l|| ||\mathbf{\Phi}\mathbf{r}_i|| \cos\alpha_l | / \sum_{l=1}^{L} |\cos\alpha_l| \right) \\ &= -\sum_{l=1}^{m} \left(|\langle \mathbf{\Phi}\mathbf{e}_l, \mathbf{\Phi}\mathbf{r}_i \rangle| / \sum_{l=1}^{m} |\langle \mathbf{\Phi}\mathbf{e}_l, \mathbf{\Phi}\mathbf{r}_i \rangle| \right) \\ &\quad \times \log \left(|\langle \mathbf{\Phi}\mathbf{e}_l, \mathbf{\Phi}\mathbf{r}_i \rangle| / \sum_{l=1}^{m} |\langle \mathbf{\Phi}\mathbf{e}_l, \mathbf{\Phi}\mathbf{r}_i \rangle| \right) \\ &= -\sum_{l=1}^{m} \left(|\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi}\mathbf{r}_i| / \sum_{l=1}^{m} |\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi}\mathbf{r}_i| \right) \\ &\quad \times \log \left(|\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi}\mathbf{r}_i| / \sum_{l=1}^{m} |\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi}\mathbf{r}_i| \right) \end{aligned} \tag{A5}$$

where the norm $||\cdot||$ in (A5) is the l_2-norm, $||\cdot||_2$.

Now as $m \to L$, $\mathbf{\Phi}^T\mathbf{\Phi} \to \mathbf{I}$ and $\delta_k \to 0$ Then, we use (A5) to derive

$$\begin{aligned} H(\mathbf{\Phi}\mathbf{r}_i) &= -\sum_{l=1}^{L} \left(|\mathbf{e}_l^T \mathbf{r}_i| / \sum_{l=1}^{L} |\mathbf{e}_l^T \mathbf{r}_i| \right) \log \left(|\mathbf{e}_l^T \mathbf{r}_i| / \sum_{l=1}^{L} |\mathbf{e}_l^T \mathbf{r}_i| \right) \\ &= -\sum_{l=1}^{L} \frac{|r_{il}|}{\bar{r}_i} \log \frac{|r_{il}|}{\bar{r}_i} \\ &= -\sum_{l=1}^{L} p_{il} \log p_{il} = H(\mathbf{p}_i) \end{aligned} \tag{A6}$$

which implies that $H(\mathbf{\Phi}\mathbf{r}_i) = H(\mathbf{p}_i)$. Since $\mathbf{p}_i$ is the normalized probability vector of $\mathbf{r}_i$ resulting from (6), $H(\mathbf{\Phi}\mathbf{r}_i) = H(\mathbf{\Phi}\mathbf{r}\mathbf{p}_i)$. The theorem is proved. ■

Appendix B
Proof of Corollary

$$\begin{aligned} &H(\mathbf{\Phi}\mathbf{p}_i, \mathbf{\Phi}\mathbf{q}_j) \\ &= -\sum_{l=1}^{m} \left(|\langle \Phi\mathbf{e}_l, \mathbf{\Phi}\mathbf{p}_i \rangle| / \sum_{l=1}^{m} |\langle \mathbf{\Phi}\mathbf{e}_l, \mathbf{\Phi}\mathbf{p}_i \rangle| \right) \\ &\quad \times \log \left(|\langle \mathbf{\Phi}\mathbf{e}_l, \mathbf{\Phi}\mathbf{q}_j \rangle| / \sum_{l=1}^{m} |\langle \mathbf{\Phi}\mathbf{e}_l, \mathbf{\Phi}\mathbf{q}_j \rangle| \right) \\ &= -\sum_{l=1}^{m} \left(|\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi}\mathbf{p}_i| / \sum_{l=1}^{m} |\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi}\mathbf{p}_i| \right) \\ &\quad \times \log \left(|\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi} q_j| / \sum_{l=1}^{m} |\mathbf{e}_l^T \mathbf{\Phi}^T \mathbf{\Phi}\mathbf{q}_j| \right). \end{aligned} \tag{A7}$$

Since $\mathbf{\Phi}^T\mathbf{\Phi} \to \mathbf{I}$ as $m \to L$, (A7) becomes $H(\mathbf{p}_i, \mathbf{q}_j)$ which results in (10).

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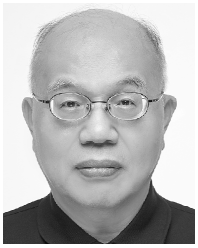
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Bernard Lampe (Student Member, IEEE) received the bachelor's degree in computer science from the University of Michigan, Ann Arbor, MI, USA, in 2009 and the master's and Ph.D. degrees in electrical engineering from the University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, in 2016 and 2019, respectively.

He is currently a Senior Principal Vulnerability Researcher with the Raytheon Cyber Solutions, Inc., Melbourne, FL, USA. His research areas are software vulnerability discovery and exploitation in addition to merging hyperspectral image processing and compressive sensing technologies.



Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from the Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, in 1982, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

Since 1987, he has been with the University of Maryland, Baltimore County (UMBC), Baltimore, MD, where he is currently a Professor with the Department of Computer Science and Electrical Engineering. He has been holding the Chang Jiang Scholar Chair Professorship and the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, since 2016. He has also been the Chair Professor of National Chiao Tung University, Hsinchu, Taiwan, since 2019. He has authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic, 2003) and *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection* (Springer, 2016), and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017). He has edited two books, *Recent Advances in Hyperspectral Signal and Image Processing* (2006) and *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007) and co-edited with A. Plaza the book *High Performance Computing in Remote Sensing* (CRC Press, 2007). He has seven patents on hyperspectral image processing. His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE. He received a National Research Council Senior Research Associateship Award from 2002 to 2003 sponsored by the U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, MD, USA. He was the Guest Editor of a Special Issue of the *Journal of High Speed Networks* on Telemedicine and Applications in April 2000 and Co-Guest Editor of another special issue of the same journal on Broadband Multimedia Sensor Networks in Healthcare Applications in April 2007. He was also the Co-Guest Editor of special issues on High Performance Computing of Hyperspectral Imaging for the *International Journal of High Performance Computing Applications* in December 2007, Signal Processing and System Design in Health Care Applications for the *EURASIP Journal on Advances in Signal Processing* in 2009, Multipsectral, Hyperspectral, and Polarimetric Imaging Technology for *Journal of Sensors* in 2016, and Hyperspectral Imaging and Applications for *Remote Sensing* in 2018. He was a Plenary Speaker for Society for Photo-optical Instrumentation Engineers (SPIE) Optics+Applications, Remote Sensing Symposium in 2009.

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Adam Bekit (Student Member, IEEE) received the B.S. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1997 and the M.S. and Ph.D. degrees in electrical engineering from the University of Maryland, Baltimore County (UMBC), Baltimore, MD, in 2017 and 2019, respectively.

He is a member of the Remote Sensing, Signal, and Image Processing Laboratory with the UMBC. His research interests include hyperspectral imaging, remote sensing, compressive sensing, signal processing, and machine learning.



Charles Della Porta (Student Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the University of Central Florida, Orlando, FL, USA, in 2007 and 2011, respectively, and the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore County, Baltimore, MD, USA, in 2019.

He is currently a Senior Physicist with The Johns Hopkins University Applied Physics Laboratory, Laurel, MD, USA. His research interests include signal processing and machine learning, with a particular focus on applications in acoustics, electromagnetics, and hyperspectral imaging.

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